

Part 1: Your Hypotheses

Before running any experiments, predict the rejection rate for each defense configuration against each attack type. Write a percentage (0% or 100%) in each cell.

Experiment	Immediate Replay	Delayed Replay	Modified Replay
No defense	0%	0%	0%
Timestamp only	0%	100%	0%
Counter only	100%	100%	0%
All defenses	100%	100%	100%

Part 2: Experiment Results

Experiment 1: No Defense (Baseline)

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	Yes 5	0	0%
Delayed replay	Yes 5	0	0%
Modified replay	Yes 5	0	0%

Experiment 2: Timestamp Only

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	Yes 5	0	0%
Delayed replay	0	5	100%
Modified replay	Yes 5	0	0%

Experiment 3: Counter Only

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	0	Yes 5	100%
Delayed replay	0	Yes 5	100%
Modified replay	Yes 5	0	0%

Experiment 4: All Defenses

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	0	Yes 5	100%
Delayed replay	0	Yes 5	100%
Modified replay	0	Yes 5	100%

Part 3: Summary Table

Compile all your results into a single table:

Defense	Immediate Replay	Delayed Replay	Modified Replay
No defense	0%	0%	0%
Timestamp only	0%	100%	0%
Counter only	100%	100%	0%
All defenses	100%	100%	100%

Part 4: Screenshots

Terminal output (experiment summary):

No Defense (Baseline) for replay attack experiment result:

```
Windows PowerShell
[ACCEPTED] seq=4 - No defenses active
[ACCEPTED] seq=5 - No defenses active

[STEP 3] Creating modified replay attack...
Replaying 5 MODIFIED messages

[STEP 4] Running attack...
[ACCEPTED] seq=6 - No defenses active
[ACCEPTED] seq=7 - No defenses active
[ACCEPTED] seq=8 - No defenses active
[ACCEPTED] seq=9 - No defenses active
[ACCEPTED] seq=10 - No defenses active

[RESULTS]
Messages tested: 5
Accepted: 5
Rejected: 0
Rejection rate: 0%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
none         immediate   5         0         0%
none         delayed    5         0         0%
none         modified   5         0         0%

[SAVED] Results saved to experiment_results.json
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk6\hydroficient-project>
```

Timestamp for replay attack experiment result:

```
Windows PowerShell
Message 5: seq=5, flow=49.81 LPM

[STEP 2] Processing legitimate messages through subscriber...
[ACCEPTED] seq=1 - All checks passed
[ACCEPTED] seq=2 - All checks passed
[ACCEPTED] seq=3 - All checks passed
[ACCEPTED] seq=4 - All checks passed
[ACCEPTED] seq=5 - All checks passed

[STEP 3] Creating modified replay attack...
Replaying 5 MODIFIED messages

[STEP 4] Running attack...
[ACCEPTED] seq=6 - All checks passed
[ACCEPTED] seq=7 - All checks passed
[ACCEPTED] seq=8 - All checks passed
[ACCEPTED] seq=9 - All checks passed
[ACCEPTED] seq=10 - All checks passed

[RESULTS]
Messages tested: 5
Accepted: 5
Rejected: 0
Rejection rate: 0%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
timestamp    immediate   5          0         0%
timestamp    delayed     0          5        100%
timestamp    modified    5          0         0%

[SAVED] Results saved to experiment_results.json
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk6\hydroficient-project>
```

Counter for replay attack experiment result:

```
Windows PowerShell
Message 5: seq=5, flow=52.82 LPM

[STEP 2] Processing legitimate messages through subscriber...
[ACCEPTED] seq=1 - All checks passed
[ACCEPTED] seq=2 - All checks passed
[ACCEPTED] seq=3 - All checks passed
[ACCEPTED] seq=4 - All checks passed
[ACCEPTED] seq=5 - All checks passed

[STEP 3] Creating modified replay attack...
Replaying 5 MODIFIED messages

[STEP 4] Running attack...
[ACCEPTED] seq=6 - All checks passed
[ACCEPTED] seq=7 - All checks passed
[ACCEPTED] seq=8 - All checks passed
[ACCEPTED] seq=9 - All checks passed
[ACCEPTED] seq=10 - All checks passed

[RESULTS]
Messages tested: 5
Accepted: 5
Rejected: 0
Rejection rate: 0%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
counter       immediate   0          5        100%
counter       delayed     0          5        100%
counter       modified    5          0         0%

[SAVED] Results saved to experiment_results.json
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk6\hydroficient-project>
```

All Defenses for replay attack experiment result:

```
Windows PowerShell
Message 5: seq=5, flow=46.76 LPM

[STEP 2] Processing legitimate messages through subscriber...
[ACCEPTED] seq=1 - All checks passed
[ACCEPTED] seq=2 - All checks passed
[ACCEPTED] seq=3 - All checks passed
[ACCEPTED] seq=4 - All checks passed
[ACCEPTED] seq=5 - All checks passed

[STEP 3] Creating modified replay attack...
Replaying 5 MODIFIED messages

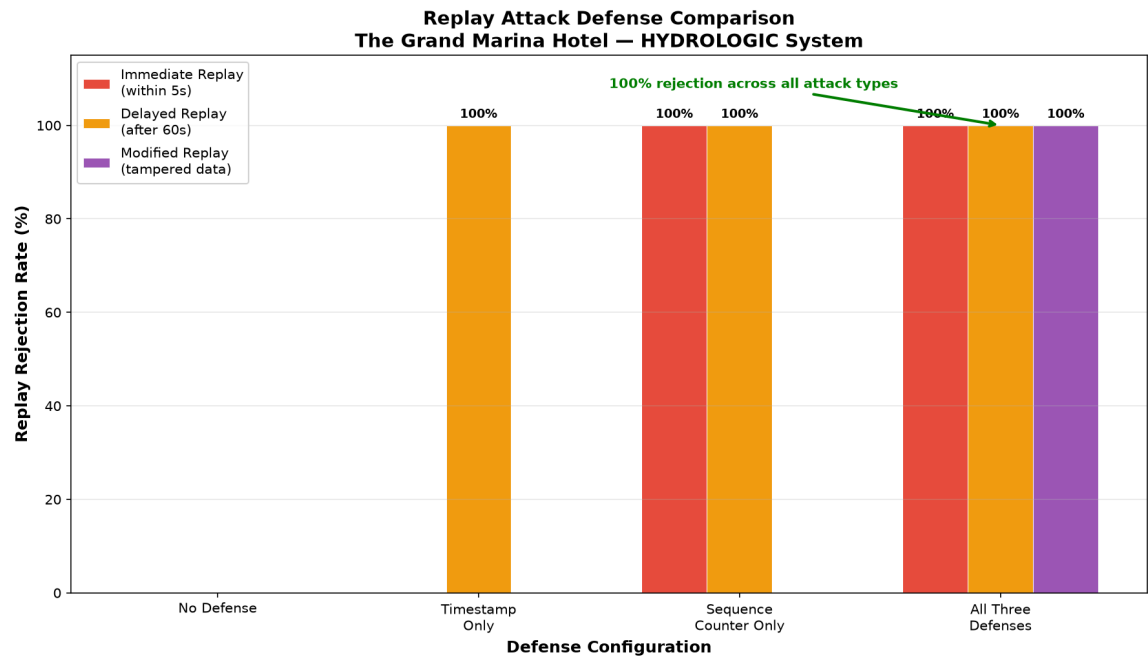
[STEP 4] Running attack...
[REJECTED] seq=6 - HMAC mismatch
[REJECTED] seq=7 - HMAC mismatch
[REJECTED] seq=8 - HMAC mismatch
[REJECTED] seq=9 - HMAC mismatch
[REJECTED] seq=10 - HMAC mismatch

[RESULTS]
Messages tested: 5
Accepted: 0
Rejected: 5
Rejection rate: 100%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
all          immediate    0         5        100%
all          delayed      0         5        100%
all          modified     0         5        100%

[SAVED] Results saved to experiment_results.json
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk6\hydroficient-project> |
```

defense_comparison.png chart:



Part 5: Hypothesis Comparison

1. Which predictions were correct?

All predictions are correct.

2. Which results surprised you? Were there any attack/defense combinations you expected to block that didn't, or vice versa?

The result does not surprise me since I read through all your instructions and understood all the logic behind each defense.

3. Why did individual defenses have gaps? Think about what each defense checks and what it doesn't.

For individual defenses, after we close the program and restart to run the subscriber and publisher, everything will count as the beginning. If the attackers copy our record and try to replay another day in the early time. The defense won't work anymore. That is the gap in the real situation.